THE UNIVERSITY OF MELBOURNE

Student number

Semester 2 Assessment, 2024

School of Mathematics and Statistics

MAST20026 Real Analysis

Reading time: 15 minutes — Writing time: 3 hours

This exam consists of 29 pages (including this page) with 10 questions and 100 total marks

Permitted Materials

- No books, notes or other printed or handwritten material are permitted.
- No calculators are permitted.
- Any mobile phones or internet-enabled devices brought into the exam room must be **turned off** and placed on the floor under your table.

Instructions to Students

- Write your answers in the boxes provided on the exam. There is extra space you can use for answers to any question commencing on page 26. If you still do not have enough space, tick the box near the bottom of page 29 and request extra pages from an invigilator—include the question number at the top of each additional page.
- You must NOT remove this question paper, or any extra pages provided to you, at the conclusion of the examination.

Instructions to Invigilators

- Students are to write their answers on the paper. They may request extra pages if they run out of space on the exam paper.
- This exam paper contains examinable material and must be collected, together with extra pages if any, at the conclusion of the examination.

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Question 1 (8 marks)

Let p and q be primitive statements.

(a) Write a truth table for the compound statement

$$(\sim p) \iff (p \vee q).$$

(b) If $(\sim p) \iff (p \vee q)$ is true, what can you conclude about the truth value of p? Justify your answer in terms of specific lines in truth table from Part (a).

(c) Using a truth table, show that $(\sim p) \iff (p \vee q)$ is logically equivalent to $\sim (q \implies p)$.

(a)

p	q	$\sim p$	$p \vee q$	$(\sim p) \iff (p \vee q)$
T	T	F	T	F
T	F	F	T	F
F	T	T	T	T
F	F	T	F	F

Correct truth table.

(b) By Line 3 of the truth table from Part (a), if $(\sim p) \iff (p \vee q)$ is true, then p must be False.

Correct answer that p must be False.

Correct justification that Line 3 refers to the situation where $(\sim p) \iff (p \vee q)$ holds true.

(c) Using Part (a)

p	q	$q \implies p$	$\sim (q \implies p)$	$(\sim p) \iff (p \vee q)$
T	T	T	F	F
T	F	T	F	F
F	T	F	T	T
F	F	T	F	F

p	q	$[(\sim p) \iff (p \vee q)] \iff [\sim (q \implies p)]$
T	T	T
T	F	T
F	T	T
F	F	T

Since $[(\sim p) \iff (p \vee q)] \iff [\sim (q \implies p)]$ is a tautology, $(\sim p) \iff (p \vee q)$ is logically equivalent to $\sim (q \implies p)$.

Correct truth table for $[(\sim p) \iff (p \vee q)] \iff [\sim (q \implies p)]$ (possibly using Part (a)).

Asserts that $[(\sim p) \iff (p \vee q)] \iff [\sim (q \implies p)]$ is a tautology.

Deduces that $[(\sim p) \iff (p \vee q)] \equiv [\sim (q \implies p)]$.

Question 2 (9 marks)

Find the largest positive integer d such that for every $n \in \mathbb{N}$, $5^n - 8n - 1$ is divisible by d. Use induction to prove your assertion is true.

(Hint: Start by writing out $5^n - 8n - 1$ for $n = 0, 1, 2, 3$ and making a reasonable guess for the value of d.)

Scrap work. Observe that

$$5^0 - 8(0) - 1 = 0, \qquad 5^1 - 8(1) - 1 = -4,$$
$$5^2 - 8(2) - 1 = 8, \qquad 5^3 - 8(3) - 1 = 100.$$

Hence we should choose $d = 4$. (Notice that since $5^1 - 8(1) - 1 = -4$, no integer larger than 4 divides $5^n - 8n - 1$ for all $n \in \mathbb{N}^+$.)

Correct answer $d = 4$.

Note: If students guess $d = 2$, the proof is the same with obvious modifications. Deduct 1 mark for the answer for d, and mark the rest of the question for a correct proof by induction with the student's chosen d.

Proof. We proceed by induction on n. We want to show that for each $n \in \mathbb{N}$ the condition

$$p(n) \ : \ 4 \text{ divides } 5^n - 8n - 1$$

holds true.

Begins the proof by asserting that we use induction to prove the condition $p(n)$: "4 divides $5^n - 8n - 1$" holds true for all $n \in \mathbb{N}^+$.
Throughout the proof, $p(n)$ is the condition that "4 divides $5^n - 8n - 1$" and not the value of $5^n - 8n - 1$.

Base case. Above we showed that when $n = 0$, $5^n - 4n - 1 = 0$. Since 4 divides 0, $p(0)$ is true.
Correct proof for the base case.

Inductive step. Suppose that for some $k \in \mathbb{N}$, $p(k)$ is true. That is, suppose that 4 divides $5^k - 8k - 1$. This means that there exists $m \in \mathbb{Z}$ such that

$$5^k - 8k - 1 = 4m.$$

We want to show that 4 divides $5^{k+1} - 8(k+1) - 1$.

We compute that

$$[5^{k+1} - 8(k+1) - 1] - [5^k - 8k - 1] = 4 \cdot 5^k - 8k = 4(5^k - 2k).$$

Hence

$$\begin{aligned} 5^{k+1} - 8(k+1) - 1 &= [5^k - 8k - 1] + 4(5^k - 2k) \\ &= 4m + 4(5^k - 2k) \\ &= 4(m + 5^k - 2k). \end{aligned}$$

Therefore, 4 divides $5^{k+1} - 8(k+1) - 1$. That is, $p(k+1)$ is true.

Alternative approach: Substitute $5^k = 4m + 8k + 1$ *into* $5 \cdot 5^k - 8(k+1) - 1$ *to get* $5^{k+1} - 8(k+1) - 1 = 4(5m + 8k - 1)$.

Correctly states the premise that "for some $k \in \mathbb{N}$, $p(k)$ is true", or similar.
Correct proof for the induction step.

Conclusion. By the Principle of Mathematical Induction, $5^n - 8n - 1$ is divisible by 4 for all $n \in \mathbb{N}$.

Correctly asserts the conclusion that "By the Principle of Mathematical Induction, $5^n - 8n - 1$ is divisible by 4 for all $n \in \mathbb{N}$.".

Question 3 (7 marks)

Listed below are the Axioms of the Real Numbers.

O1. $\forall x, y \in \mathbb{R}$ exactly one of the following is true: $x < y$, $y < x$, or $x = y$.
O2. $\forall x, y, z \in \mathbb{R}$ if $x < y$ and $y < z$ then $x < z$.
A1. $\forall x, y \in \mathbb{R} \quad x + y \in \mathbb{R}$
A2. $\forall x, y \in \mathbb{R} \quad x + y = y + x$
A3. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$
A4. $\exists 0 \in \mathbb{R} \quad \forall x \in \mathbb{R} \quad x + 0 = x$
A5. $\forall x \in \mathbb{R} \quad \exists -x \in \mathbb{R} \quad x + (-x) = 0$
M1. $\forall x, y \in \mathbb{R} \quad xy \in \mathbb{R}$
M2. $\forall x, y \in \mathbb{R} \quad xy = yx$
M3. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$
M4. $\exists 1 \in \mathbb{R} \setminus \{0\} \quad \forall x \in \mathbb{R} \quad 1x = x$
M5. $\forall x \in \mathbb{R} \setminus \{0\} \quad \exists 1/x \in \mathbb{R} \setminus \{0\} \quad x \cdot 1/x = 1$
D. $\forall x, y, z \in \mathbb{R} \quad x(y + z) = xy + xz$
OA. $\forall x, y, z \in \mathbb{R}$ if $x < y$ then $x + z < y + z$
OM. $\forall x, y \in \mathbb{R}$ if $x > 0$ and $y > 0$ then $xy > 0$
C. Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Using only the Axioms of the Real Numbers (and rules of logic), write a formal proof showing that for all non-zero $a \in \mathbb{R}$

$$1/(1/a) = a.$$

Proof.

1. Let $a \in \mathbb{R} \setminus \{0\}$.	(premise)
2. $(\exists\, 1/a \in \mathbb{R} \setminus \{0\})\ \ a \cdot 1/a = 1$	(M5)
3. $(\exists\, 1/(1/a) \in \mathbb{R} \setminus \{0\})\ \ (1/a) \cdot (1/(1/a)) = 1$	(A5)
4. $a \cdot [(1/a) \cdot (1/(1/a))] = a \cdot 1$	(3)
5. $[a \cdot (1/a)] \cdot (1/(1/a)) = a \cdot 1$	(4,M2)
6. $1 \cdot (1/(1/a)) = a \cdot 1$	(2,5)
7. $1 \cdot (1/(1/a)) = 1 \cdot a$	(6,M2)
8. $1/(1/a) = a$	(7,M4)

Proof is of the form of a formal proof.
Overall correct proof (separate from the justification).
Correct justification for each step using the Axioms of the Real Numbers.
Including using the Axioms carefully.

Note. *Here and for the remainder of the exam, you may write informal proofs, and you may assume that algebra and inequalities with the real numbers behave the way you expect.*

Question 4 (9 marks)

For each of the following situations, provide a specific example for which the situation is true, and briefly justify your answer.

(a) An unbounded set E such that the set of all limit points of E is $[0, 1]$.

(b) A non-constant sequence (f_n) which is contractive and converges to -1.

(c) A function $g : \mathbb{R} \to \mathbb{R}$ such that $|g|$ is continuous on $\mathbb{R}$ but g is not continuous at a for all $a \in \mathbb{R}$.

(a) Let $E = [0, 1] \cup \mathbb{Z}$.

E is unbounded since $\mathbb{Z} \subset E$ and $\mathbb{Z}$ is unbounded by the Archimedean Principle.

Let $a \in [0, 1)$. For each $0 < \delta \leq 1 - a$, $x = a + \delta/2$ satisfies $0 \leq a < x = a + \delta/2 < a + \delta \leq 1$ and thus $0 < |x - a| < \delta$ and $x \in [0, 1] \subseteq E$. If instead $\delta > 1 - a$, then $x = a + (1 - a)/2$ satisfies $0 < |x - a| < 1 - a < \delta$ and $x \in E$. By similar reasoning, when $a = 1$ and $0 < \delta \leq 1$, $x = 1 - \delta/2$ satisfies $0 < |x - a| < \delta$ and $x \in E$. When $a = 1$ and $\delta \geq 1$, $x = 1/2$ satisfies $0 < |x - a| < \delta$ and $x \in E$. Therefore, each $a \in [0, 1]$ is a limit point of E.

Let $a \in (-\infty, 0) \cup (1, \infty)$. If $a \notin \mathbb{Z}$, then there exists $\delta > 0$ such that $E \cap (a - \delta, a + \delta) = \emptyset$. If $a \in \mathbb{Z}$, then for $\delta = 1/2$ we have that $E \cap (a - \delta, a + \delta) = \{a\}$. Therefore, each $a \in (-\infty, 0) \cup (1, \infty)$ is not a limit point of E.

Any correct answer for the set E.

Correct justification. Must explain why E is unbounded, each point of $[0, 1]$ is a limit point of E, and why there are no other limit points of E.

(b) Let $f_n = -1 + 2^{-n}$ for each $n \in \mathbb{N}^+$.

Then

$$|f_n - f_{n+1}| = |(-1 + 2^{-n}) - (-1 + 2^{-n+1})| = |2^{-n}| = 2^{-n}$$

Thus

$$|f_n - f_{n+1}| = 2^{-n} = \frac{1}{2}|f_{n-1} - f_n|.$$

Therefore, (f_n) is contractive.

Let $\epsilon > 0$. By the Archimedean Principle, there exists $M \in \mathbb{N}^+$ such that $1/M < \epsilon$. Hence if $n > M$ then

$$|f_n - (-1)| = |(-1 + 2^{-n}) - (-1)| = |2^{-n}| = 2^{-n} \leq \tfrac{1}{n} < \tfrac{1}{M} < \epsilon.$$

Therefore, $\lim_{n\to\infty} f_n = -1$.

Any correct answer for the sequence (f_n).

Correct justification for why (f_n) is contractive.

Correct justification for why (f_n) converges to -1.

(c) Let

$$g(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ -1 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Then $|g|$ is the constant function equal to one. Since constant functions are continuous, $|g|$ is continuous.

Let $a \in \mathbb{R}$. When $a \in \mathbb{Q}$, for each $\delta > 0$ there exists $x \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < x < a + \delta$ but $|g(x) - g(a)| = |-1 - 1| = 2$. Similarly, when $a \in \mathbb{R} \setminus \mathbb{Q}$, for each $\delta > 0$ there exists $x \in \mathbb{Q}$ such that $a < x < a + \delta$ but $|g(x) - g(a)| = |1 - (-1)| = 2$. Therefore, for each $\delta > 0$ there exists $x \in \mathbb{R}$ such that $|x - a| < \delta$ but $|g(x) - g(a) \geq 2$. Therefore, g is not continuous at a.

Any correct answer for the function g. (Typically a "weird" function.)

Correct justification for why $|g|$ is continuous at each point of $\mathbb{R}$.

Correct justification for why g is not continuous at every point of $\mathbb{R}$. (Typically using $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ being dense in $\mathbb{R}$.)

Question 5 (12 marks)

Let $A, B \subset \mathbb{R}$ be sets which are both bounded above in $\mathbb{R}$. Let α be the least upper bound of A and β be the least upper bound of B.

(a) Show that $\max\{\alpha, \beta\}$ is an upper bound of the set $A \cup B$.

(b) Show that $\sup(A \cup B) = \max\{\alpha, \beta\}$.

(c) Is it always true that $\max\{\alpha, \beta\}$ is a limit point of $A \cup B$? Justify your answer with a proof or counter-example.

(a) Let $A, B \subset \mathbb{R}$ be bounded above. Let $\alpha = \sup A$ and $\beta = \sup B$. We claim that $\max\{\alpha, \beta\}$ is an upper bound of $A \cup B$.

Let $x \in A \cup B$. Then either $x \in A$ or $x \in B$. In the case that $x \in A$, since α is an upper bound of A,

$$x \leq \alpha \leq \max\{\alpha, \beta\}.$$

Similarly, in the case that $x \in B$, since β is an upper bound of B

$$x \leq \beta \leq \max\{\alpha, \beta\}.$$

Therefore, for all $x \in A \cup B$ we have that $x \leq \max\{\alpha, \beta\}$. Therefore, $A \cup B$ is bounded above, and $\max\{\alpha, \beta\}$ is an upper bound of $A \cup B$.

Argues that $\max\{\alpha, \beta\}$ is an upper bound by showing that for all $x \in A \cup B$, $x \leq \max\{\alpha, \beta\}$.

Correct argument that if $x \in A \cup B$, then $x \leq \max\{\alpha, \beta\}$.

(b) Let $A, B \subset \mathbb{R}$ be bounded above. Let $\alpha = \sup A$ and $\beta = \sup B$. Without loss of generality, assume that $\alpha \leq \beta$ and note that $\beta = \max\{\alpha, \beta\}$. We claim that β is the *least* upper bound of $A \cup B$.

Let $\gamma \in \mathbb{R}$ such that $\gamma < \beta$. Since $\beta = \sup B$ and $\gamma < \beta$, γ cannot be an upper bound of B. Thus there exists $x \in B$ such that $x > \gamma$. Since $x \in B$, $x \in A \cup B$. Therefore, there exists $x \in A \cup B$ such that $x > \gamma$. Therefore, γ is not an upper bound of $A \cup B$.

Since β is an upper bound of $A \cup B$ and there is no upper bound γ of $A \cup B$ such that $\gamma < \beta$, β is the least upper bound of $A \cup B$.

Argues that $\max\{\alpha, \beta\}$ is not a least upper bound by attempting to show any one of:

- A direct proof that if $\gamma < \max\{\alpha, \beta\}$ then γ is not an upper bound of $A \cup B$.
- A proof by contradiction that if γ is an upper bound of $A \cup B$ and $\gamma < \max\{\alpha, \beta\}$, then we reach a contradiction.
- A proof that for every $\epsilon > 0$ there exists $x \in A \cup B$ with $x > \max\{\alpha, \beta\} - \epsilon$.

A correct argument that $\max\{\alpha, \beta\}$ is not a least upper bound.

In Parts (a) and (b), correctly states the premise and conclusions of the proofs.

(c) No. Let $A = \{1, 3\}$ and $B = \{2, 3\}$. Then $A \cup B = \{1, 2, 3\}$. Since $A \cup B$ are finite sets and the largest element of each set is 3, $\sup A = 3$ and $\sup B = 3$. However, for $\delta = 1/2 > 0$, there is no element $x \in A \cup B$ such that $0 < |x - 3| < \delta$. Therefore, 3 is not a limit point of $A \cup B$.

Correct answer No.

Any correct specific counter-example for A and B (namely, the maximum of $A \cup B$ is an isolated point of $A \cup B$).

Correctly explains what the counter-example is valid.

Question 6 (10 marks)

Consider the sequence (f_n) defined by

$$f_n = \sqrt{2n-1}$$

for each $n \in \mathbb{N}^+$.

(a) Show that (f_n) is a monotone increasing sequence.

(b) Use the $\epsilon - M$ definition of convergence to show that

$$\lim_{n\to\infty} |f_n - f_{n+1}| = 0.$$

(c) Is the sequence (f_n) Cauchy? Justify your answer using definitions and theorems from lecture.

(a) For each $n \in \mathbb{N}^+$, using the monotonicity of square roots

$$\begin{aligned} n &< n+1 \\ 2n-1 &< 2(n+1)-1\sqrt{2n-1} \qquad < \sqrt{2(n+1)-1} \\ f_n &< f_{n+1} \end{aligned}$$

Therefore, (f_n) is monotone increasing.

Argues that the definition of monotone increasing is true, i.e. that $f_n \leq f_{n+1}$ for all $n \in \mathbb{N}^+$.

Correct algebraic argument showing that $f_n < f_{n+1}$ for all $n \in \mathbb{N}^+$.

(b) Let $\epsilon > 0$. We want to show that there exists $M \in \mathbb{N}^+$ such that

$$n > M \implies |f_n - f_{n+1}| < \epsilon.$$

We compute that

$$\begin{aligned} |f_n - f_{n+1}| &= \left|\sqrt{2n-1} - \sqrt{2n+1}\right| \\ &= \sqrt{2n+1} - \sqrt{2n-1} \\ &= \frac{2}{\sqrt{(2n+1)(2n-1)}} = \frac{2}{\sqrt{4n^2-1}}. \end{aligned}$$

There's an error here. The denominator should be \sqrt(2n+1) + \sqrt(2n-1).
That means the subsequent epsilon-delta proof isn't for the correct expression.

Let $M = \lceil \sqrt{2/\epsilon^2 + 1/2} \rceil$. If $n > M$ then

$$\begin{aligned} n &> \sqrt{\frac{2}{\epsilon^2} + \frac{1}{2}} \\ n^2 &> \frac{2}{\epsilon^2} + \frac{1}{2} \\ \frac{n^2}{2} - \frac{1}{4} &> \frac{1}{\epsilon^2} \\ \frac{4}{4n^2 - 1} &< \epsilon^2 \\ |f_n - f_{n+1}| \leq \frac{2}{\sqrt{4n^2 - 1}} &< \epsilon \end{aligned}$$

Therefore, if $n > M$, then $|f_n - f_{n+1} - 0| < \epsilon$. Therefore, $\lim_{n\to\infty} |f_n - f_{n+1}| = 0$.

Argues via an $\epsilon - M$ proof. That is, assumes $\epsilon > 0$ and chooses $M \in \mathbb{N}^+$. Then argues that if $n > M$ then $|f_n - f_{n+1} - 0| < \epsilon$.

Any correct value for M.

Correct algebraic argument showing that if $n > M$ then $|f_n - f_{n+1}| < \epsilon$.

(c) No, since (f_n) is not bounded above, (f_n) is not Cauchy.

To see that (f_n) is not bounded above, let $r \in \mathbb{R}$. By the Archimedean Principle, there exists $n \in \mathbb{N}^+$ such that $n > (r^2+1)/2$. Hence

$$\begin{aligned} n &> \frac{r^2 + 1}{2} \\ 2n - 1 &> r^2 \\ f_n = \sqrt{2n - 1} &> r \end{aligned}$$

Correct answer that (f_n) is *not* Cauchy.

Any complete and correct explanation for why (f_n) is not Cauchy.

Question 7 (11 marks)

Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 3x - 2 & \text{if } x \in \mathbb{Q} \\ x^3 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

(a) Using the $\epsilon - \delta$ definition of continuity, show that f is not continuous at 0.

(b) Using an $\epsilon - \delta$ argument, show that f is differentiable at $x = 1$.

(a) Let $\epsilon = 1$. Let $\delta > 0$. Let x be an irrational number such that $0 < x < \delta$. Then

$$|f(x) - f(0)| = |x^3 - (3 \cdot 0 - 2)| = |x^3 + 2| = x^3 + 2 \geq 1.$$

Therefore, for every $\delta > 0$ there exists $x \in \mathbb{R}$ such that $|x - 0| < \delta$ and $|f(x) - f(0)| \geq 1$. Therefore, f is not continuous at 0.

Any correct value for ϵ.

Any correct value for x.

A correct argument that when δ is small, $|f(x) - f(0)| \geq \epsilon$.

(b) We want to show that

$$\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1} = 3.$$

Let $\epsilon > 0$. Let $\delta = \min\{1, \epsilon/4\}$. Let $x \in \mathbb{R}$ such that $0 < |x - 1| < \delta$. If $x \in \mathbb{Q}$, then

$$\begin{aligned} \left| \frac{f(x) - f(1)}{x - 1} - 3 \right| &= \left| \frac{(3x - 2) - (3 \cdot 1 - 2)}{x - 1} - 3 \right| \\ &= |3 - 3| = |0| = 0 < \epsilon. \end{aligned}$$

If $x \notin \mathbb{Q}$, then

$$\begin{aligned}\left|\frac{f(x)-f(1)}{x-1}-3\right| &= \left|\frac{x^3-(3\cdot 1-2)}{x-1}-3\right| = \left|\frac{x^3-1}{x-1}-3\right| \\ &= |(x^2+x+1)-3| = |x^2+x-2| \\ &= |x+2|\,|x-1|.\end{aligned}$$

Since $|x-1| < \delta \leq 1$, by the triangle inequality

$$|x+2| \leq |x-1| + |3| \leq 1+3 = 4.$$

Thus, since $|x-1| < \delta \leq \epsilon/4$,

$$\left|\frac{f(x)-f(1)}{x-1}-3\right| = |x+2|\,|x-1| \leq 4|x-1| < 4\cdot\frac{\epsilon}{4} = \epsilon.$$

Therefore, if $|x-1| < \delta$ then $\left|\frac{f(x)-f(1)}{x-1}-3\right| < \epsilon$. In other words,

$$\lim_{x\to 1}\frac{f(x)-f(1)}{x-1} = 3.$$

Therefore, f is differentiable at 1 with $f'(1) = 3$.

Correctly writes the $\epsilon-\delta$ definition of "$f'(1) = 3$".

Any correct value of δ.

Correct argument for x rational.

Correctly bounds $|x+2|$ when x is irrational.

A correct computation when x is irrational that if $0 < |x-1| < \delta$, then $\left|\frac{f(x)-f(1)}{x-1}-3\right| < \epsilon$.

In Parts (a) and (b), correctly states the premise and conclusion. (Do not assess of the choice of parameters, e.g. the choice of ϵ in Part (a).)

Question 8 (12 marks)

Let $f : [-1, 1] \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x+1 & \text{if } -1 \le x < 0 \\ 0 & \text{if } x = 0 \\ x-1 & \text{if } 0 < x \le 1 \end{cases}$$

Let $n \in \mathbb{N}^+$ and let P be the partition

$$P = \left\{-1, -\tfrac{n-1}{n}, \ldots, -\tfrac{2}{n}, -\tfrac{1}{n}, 0, \tfrac{1}{n}, \tfrac{2}{n}, \ldots, \tfrac{n-1}{n}, 1\right\}$$

(so that $x_i = -1 + i/n$ for each i).

(a) Draw a picture of illustrating how to compute the lower Riemann sum $L(f, P)$ when $n = 4$. Include the graph of f, the partition P of $[-1, 1]$, and the rectangles for each partition. The sum of the area of these rectangles should equal $L(f, P)$.

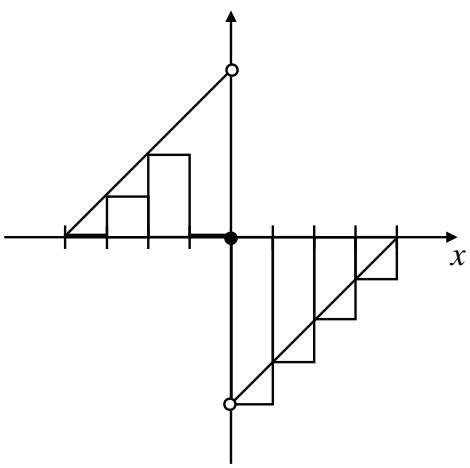
(b) Let $n \in \mathbb{N}^+$. Compute the lower and upper Riemann sums with respect to the partition

$$P = \left\{-1, -\tfrac{n-1}{n}, \ldots, -\tfrac{2}{n}, -\tfrac{1}{n}, 0, \tfrac{1}{n}, \tfrac{2}{n}, \ldots, \tfrac{n-1}{n}, 1\right\}$$

(so that $x_i = -1 + i/n$ for each i).

(c) Is f Riemann integrable on $[-1, 1]$? Explain.

(a)



Correctly sketches the graph of f.

Correctly sketches the rectangles.

(b) The i-th subinterval is $I_i = \left[-1 + \frac{i-1}{n}, -1 + \frac{i}{n}\right]$. We compute that

$$m_i = \inf_{x \in I_i} f(x) = f\left(-1 + \tfrac{i-1}{n}\right) = \tfrac{i-1}{n} \quad \text{for } 1 \le i < n,$$

$$m_n = \inf_{x \in [-1/n, 0]} f(x) = f(0) = 0,$$

$$m_{n+i} = \inf_{x \in I_{n+i}} f(x) = f\left(\tfrac{i-1}{n}\right) = \tfrac{i-1}{n} - 1 = \tfrac{i-1-n}{n} \quad \text{for } 1 \le i \le n.$$

Hence

$$\begin{aligned} L(f,P) &= \sum_{i=1}^{2n} m_i(x_i - x_{i-1}) \\ &= \sum_{i=1}^{n-1} \tfrac{i-1}{n} \cdot \tfrac{1}{n} + 0 + \sum_{i=1}^{n} \tfrac{i-1-n}{n} \cdot \tfrac{1}{n} \\ &= \sum_{i=1}^{n-1} \tfrac{i-1}{n} \cdot \tfrac{1}{n} + 0 + \sum_{j=1}^{n} \tfrac{-j}{n} \cdot \tfrac{1}{n} \\ &= \tfrac{-(n-1)}{n} \cdot \tfrac{1}{n} - 1 \cdot \tfrac{1}{n} = \tfrac{-2n+1}{n^2} \end{aligned}$$

For the upper Riemann sum, we compute that

$$\begin{aligned} M_i &= \inf_{x \in I_i} f(x) = f\left(-1 + \tfrac{i}{n}\right) = \tfrac{i}{n} \quad \text{for } 1 \le i \le n, \\ M_{n+1} &= \inf_{x \in [0,1/2]} f(x) = f(0) = 0 \\ M_{n+i} &= \inf_{x \in I_{n+i}} f(x) = f\left(\tfrac{i}{n}\right) = \tfrac{i}{n} - 1 = \tfrac{i-n}{n} \quad \text{for } 1 < i \le n, \end{aligned}$$

Hence

$$\begin{aligned} U(f,P) &= \sum_{i=1}^{2n} M_i(x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} \tfrac{i}{n} \cdot \tfrac{1}{n} + 0 + \sum_{i=2}^{n} \tfrac{i-n}{n} \cdot \tfrac{1}{n} \\ &= \sum_{i=1}^{n} \tfrac{i}{n} \cdot \tfrac{1}{n} + 0 + \sum_{j=0}^{n-2} \tfrac{-j}{n} \cdot \tfrac{1}{n} \\ &= \tfrac{(n-1)}{n} \cdot \tfrac{1}{n} + 1 \cdot \tfrac{1}{n} = \tfrac{2n-1}{n^2} \end{aligned}$$

Correct answer and approach for computing the lower Riemann sum $L(f,P)$.

Correct answer and approach for computing the upper Riemann sum $U(f,P)$.

(c) Yes, f is Riemann integrable with $\displaystyle\int_{-1}^{1} f(x)\,dx = 0$.

Let $\epsilon > 0$. By the Archimedean Principle, there exists $n \in \mathbb{N}^+$ such that $1/n < \epsilon/4$. Let $P = \left\{-1, -\frac{n-1}{n}, \ldots, \frac{n-1}{n}, 1\right\}$. Then

$$L(f,P) = \frac{-2n+1}{n^2} > \frac{-2}{n} > -\epsilon/2,$$
$$U(f,P) = \frac{2n-1}{n^2} < \frac{2}{n} < \epsilon/2$$

and thus $U(f,P) - L(f,P) < \epsilon$. Since for every $\epsilon > 0$ there exists a partition P such that $U(f,P) - L(f,P) < \epsilon$, by a theorem from lecture we know that f is Riemann integrable on $[-1,1]$.

Correct answer that f is integrable.

Correct argument for why f is integrable.

Note. *For Questions 9 and 10, you may assume without $\epsilon - M$ proofs that limits behave the way you expect.*

Question 9 (13 marks)

(a) For each of the following infinite series, use an appropriate test or theorem to prove that the series converges or diverges.

(i) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\log n}}$

(ii) $\displaystyle\sum_{n=1}^{\infty} 2^n e^{-n}$

(b) Find the radius of convergence and interval of convergence for the power series

$$\sum_{n=1}^{\infty} \frac{nx^n}{4^n(n+1)^2}$$

(a) (i) Diverges.

Let $f(x) = \frac{1}{x\sqrt{\log x}}$ for each $x > 1$. Then f is positive and continuous. If $1 < x < y$, then $\log x < \log y$ and thus $f(x) = \frac{1}{x\sqrt{\log x}} > \frac{1}{y\sqrt{y}} = f(y)$. Hence f is monotone decreasing function. We have that

$$\int_e^{\infty} \frac{1}{x\sqrt{\log x}}\,dx = \int_1^{\infty} \frac{1}{\sqrt{u}}\,du = \left[2\sqrt{u}\right]_1^{\infty} = +\infty.$$

By the Integral Test, the series $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\log n}}$ diverges.

Correct answer that the sum diverges.

Correct justification using a test for convergence of an infinite series (likely the integral test).

(ii) Converges.

Since $r = 2/e < 1$, by the Geometric Series Test, $\sum_{n=1}^{\infty} 2^n e^{-n}$ converges.

Correct answer that the sum diverges.

Correct justification using a test for convergence of an infinite series (likely the integral test).

(b) We have that

$$\begin{aligned}\lim_{n\to\infty}\frac{|a_{n+1}|}{|a_n|} &= \lim_{n\to\infty}\frac{(n+1)\,|x|^{n+1}}{4^{n+1}(n+2)^2}\cdot\frac{4^n(n+1)^2}{n\,|x|^n}\\ &= \lim_{n\to\infty}\frac{(n+1)^3\,|x|}{4n(n+2)^2}\\ &= \frac{|x|}{4}.\end{aligned}$$

By the Limit Ratio Test, the series converges whenever $|x|/4 < 1$, i.e. $|x| < 4$. Similarly, the series diverges whenever $|x| > 4$. Therefore, the radius of convergence is 4.

At the endpoint $x = -4$, the series is

$$\sum_{n=1}^{\infty}\frac{(-1)^n n}{(n+1)^2}$$

For each $n \in \mathbb{N}^+$

$$\begin{aligned}1 &\le n\\ n^3+3n^2+3n+1 &< n^3+4n^2+4n\\ (n+1)^3 &< n(n+2)^2\\ \frac{n+1}{(n+2)^2} &< \frac{n}{(n+1)^2}\end{aligned}$$

Since

$$\lim_{n\to\infty}\frac{n}{(n+1)^2} = 0$$

The series converges at $x = -4$.

At the endpoint $x = 4$, the series is

$$\sum_{n=1}^{\infty}\frac{n}{(n+1)^2}$$

By direct comparison with the p-series $\sum_{n=1}^{\infty}\frac{1}{n}$, which diverges, the series converges at $x = 4$.

Therefore, the interval of convergence is $[-4, 4)$.

Correctly answers that the radius of convergence is 4.

Correctly uses the Limit Ratio Test to find the radius of convergence.

Correct argument for why the series diverges at $x = -4$.

Correct argument for why the series converges at $x = -4$.

Correctly answers that the interval of convergence is $[-4, 4)$.

Throughout the question, whenever using the Integral Test or Alternating Series Test, checks that the corresponding function/sequence is monotone decreasing. (In the unlikely event that the student does not use either test, simply give the mark.)

Question 10 (9 marks)

Consider the function $f : (-8, \infty) \to \mathbb{R}$ defined by

$$f(x) = (8 + x)^{-1/3}$$

for each $x > -8$.

(a) Find the Taylor series expansion of f at 0.

You do not need to justify your formula for the coefficients by a rigorous proof by induction.

(b) Show that the Taylor series of f converges to f on the interval $[-4, 4]$.

(a) We compute that

$$\begin{aligned} f(x) &= (8+x)^{-1/3} \\ f'(x) &= \frac{-1}{3}(8+x)^{-4/3} \\ f''(x) &= \frac{1 \cdot 4}{3^2}(8+x)^{-7/3} \\ f^{(3)}(x) &= \frac{1 \cdot 4 \cdot 7}{3^3}(8+x)^{-10/3} \end{aligned}$$

From this we can infer that

$$f^{(n)}(x) = (-1)^n \frac{1 \cdot 4 \cdot 7 \cdots (3n-2)}{3^n}(8+x)^{-(3n-1)/3}$$

Evaluating at $x = 0$,

$$f^{(n)}(0) = (-1)^n \frac{1 \cdot 4 \cdot 7 \cdots (3n-2)}{3^n} \frac{1}{2^{3n-1}}$$

Hence the Taylor series of f is

$$T(x) = \sum_{n=0}^{\infty} (-1)^n \frac{1 \cdot 4 \cdot 7 \cdots (3n-2)}{3^n n!} \frac{x^n}{2^{3n-1}}.$$

Correct computation of $f^{(n)}(x)$.

Correct value for $f^{(n)}(0)$.

Correct answer for the Taylor series of f.

(b) Note that if $x = 0$, then $T_n(x) = f(0)$ for all $n \in \mathbb{N}^+$ as a constant sequence, and thus the Taylor series converges to $f(0)$.

Let $x \in [-4, 4] \setminus \{0\}$. By the Taylor Remainder Estimate Theorem, there exists c between 0 and x such that

$$|T_n(x) - f(x)| \leq \frac{|f^{(n+1)}(c)|}{(n+1)!} |x|^{n+1}.$$

Since

$$\begin{aligned}
\frac{|f^{(n+1)}(c)|}{(n+1)!} |x|^{n+1} &\leq \frac{1 \cdot 4 \cdot 7 \cdots (3n+1)}{3^{n+1}(n+1)!} (8-c)^{-(3n+2)/3} |x|^{n+1} \\
&\leq \frac{3 \cdot 6 \cdot 9 \cdots (3n+3)}{3^{n+1}(n+1)!} (8-c)^{-(3n+2)/3} |x|^{n+1} \\
&= \frac{3^{n+1}(n+1)!}{3^{n+1}(n+1)!} (8-c)^{-(3n+2)/3} |x|^{n+1} \\
&= (8-c)^{-(3n+2)/3} |x|^{n+1}
\end{aligned}$$

Note that

$$\begin{aligned}
|c| < |x| \leq 4 \quad &\implies \quad |x| + |c| < 8 \\
&\implies \quad |x| < 8 - |c| \\
&\implies \quad \frac{|x|}{8 - |c|} < 1.
\end{aligned}$$

Hence

$$\lim_{n\to\infty} |T_n(x) - f(x)| \leq \lim_{n\to\infty} (8-c)^{-(3n+2)/3} |x|^{n+1} = 0.$$

Explains that $T_n(0) = f(0)$ and thus the Taylor series converges to f at $x = 0$.

Uses and correctly states the Taylor Remainder Estimate Theorem.

Correctly bounds the error term and explains why the error term tends to zero.

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